

Practical No.2

(A). Aim: Use CrypTool to encrypt & decrypt passwords using RC4 Algorithm.

Theory:

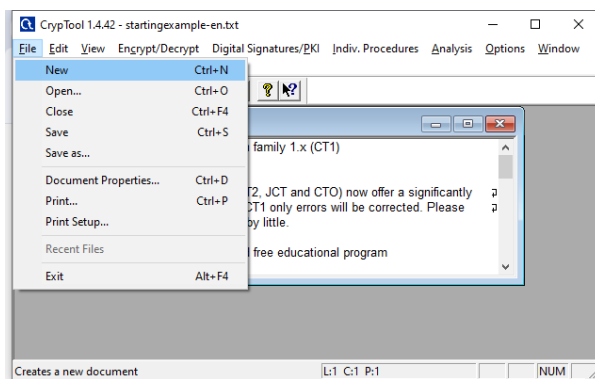
RC4 (Rivest Cipher 4) is a symmetric stream cipher algorithm used for encryption and decryption of data. It uses the same secret key for both encryption and decryption. RC4 generates a pseudo-random keystream that is combined with the plaintext using the XOR operation to produce ciphertext.

In this experiment, **CrypTool** is used to perform encryption and decryption of passwords using the RC4 algorithm. The password is converted into encrypted form (ciphertext) using a secret key. The same key is used to decrypt the ciphertext back into the original password.

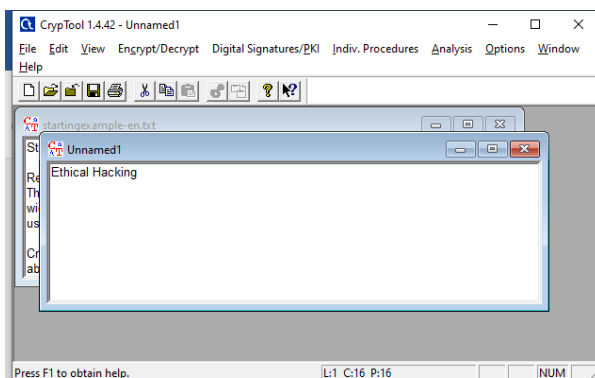
Advantages of RC4:

- Simple and fast encryption algorithm.
- Requires less memory.
- Suitable for stream-based encryption.

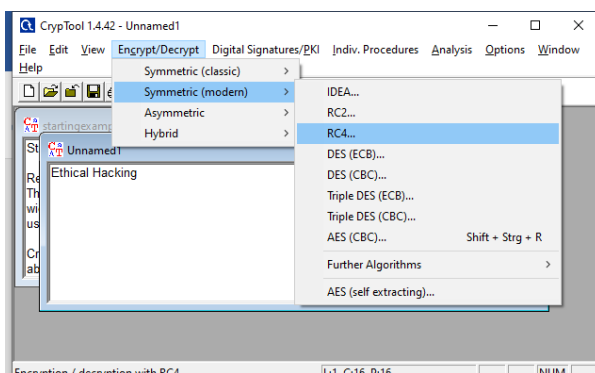
Step 1: Open CrypTool → Take a new file.



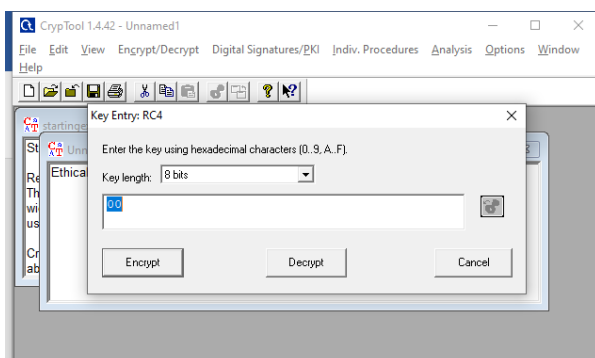
Step 2: Enter your password



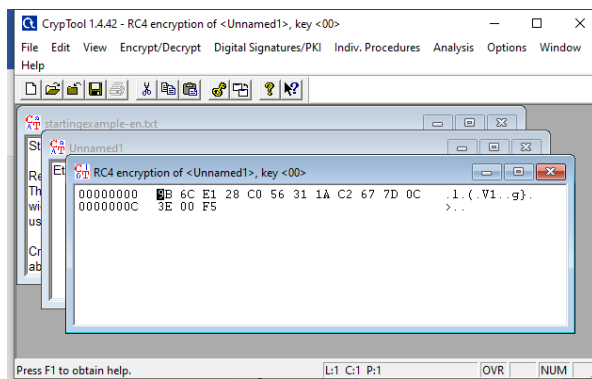
Step 3: To encrypt your password, Navigate to Encrypt/Decrypt Tab → Symmetric(Modern) → select RC4.



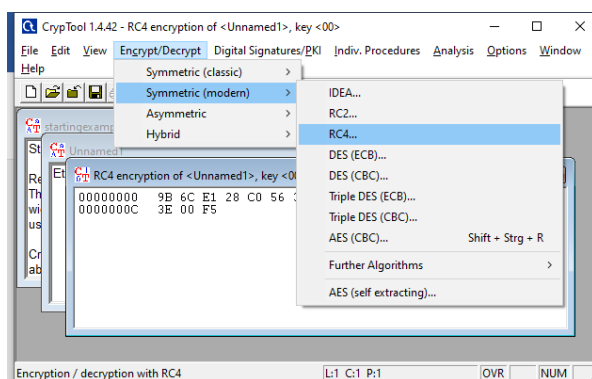
Step 4: After that click on Encrypt Button.



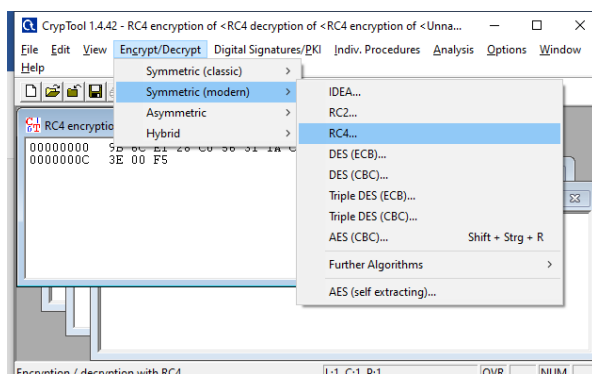
Step 5: The following window will get display.



Step 6: To decrypt your password, Navigate to Encrypt/Decrypt Tab → Symmetric(Modern) → select RC4.



Step 7: Click on Decrypt.



Step 8: your password get decrypted.

